

for m sufficiently large and divisible. Asymptotically, $W_\bullet^t$ records the hypersurfaces that appear as tangent cones of sections of L with multiplicity t at x . The challenge is to provide an intrinsic geometric interpretation of the family of graded linear series $W_\bullet^t$. When $X = C_1 \times \dots \times C_n$, intersection theory on $\mathrm{Bl}_x X$ implies that $W_\bullet^t$ is contained in a family of complete linear series associated to divisors on X_Σ , the blow-up of $\mathbb{P}^{n-1}$ in n points in general linear position. We show that this inclusion is asymptotically an equality. For this we use that X_Σ is a toric variety and the asymptotic behaviour of complete linear series on them can be managed through convex geometry. As a consequence, the vertical slices of $\Delta_x(L)$ are NOBodies associated to a ray of divisors on X_Σ with respect to a generic (strict transform) linear flag. Lastly, the global (gluing all $t \geq 0$) NObody on this special ray of divisors on X_Σ is exactly the convex set described in the statement, as we are able to prove the existence of sections whose valuation are the vertices of the simplex.

For box-products of arbitrary degrees the vertical slices of $\Delta_x(L)$ are still a family of NOBodies on X_Σ with respect to general linear flags. But it is harder to guess the vertices. It seems an interesting problem to find an algorithm that starts with a polytope in $\mathbb{R}^{n-1}$ that is a NObody on a toric variety with respect to a T -invariant flag, and construct another polytope, that will be a NObody on the same toric space but with respect to a generic linear flag.

In dimension $n \leq 3$, we can find non-trivial sections on X_Σ , whose valuations provide the vertices of a slice of the global NObody on X_Σ with respect to general linear flags. More precisely:

Theorem 1.2. *Let $X = C_1 \times \dots \times C_n$ be a product of smooth complex projective curves, polarized by a box product $L = L_1 \boxtimes \dots \boxtimes L_n$ with $\deg L_i = d_i$. Assume $d_1 \geq d_2 \geq \dots \geq d_n > 0$. Let $x \in X$ be any point.*

- *If $n = 2$, then $\Delta_x(L)$ is the isosceles trapezoid with vertices $(0, 0)$, (d_2, d_2) , $(d_1 + d_2, 0)$, (d_1, d_2) .*
- *If $n = 3$, then $\Delta_x(L)$ is the convex hull of the 9 points $(0, 0, 0)$, $(d_3, d_3, 0)$, $(d_2, d_3, d_2 - d_3)$, $(d_1, d_3, d_2 - d_3)$, $(d_1 + d_2 - d_3, d_3, 0)$, $(d_2 + d_3, 0, d_2 + d_3)$, $(d_1 + d_3, 0, d_2 + d_3)$, $(d_1 + d_2, 0, 2d_3)$, and $(d_1 + d_2 + d_3, 0, 0)$.*

In both cases, the shapes degenerate to the simplex in Theorem 1.1 when $d_i = 1$ for all i , while when the d_i are not all equal the shapes are not simplicial.

It turns out that the vertices of our iNOBodies from Theorem 1.1 and Theorem 1.2 happen exactly when there is a change in the associated base loci of the graded linear series $W_\bullet^t$. These ideas are generalized by the authors in [FL25b] on any projective variety X by connecting three aspects of local positivity of a line bundle L on X at a point $x \in X$. On one hand we have infinitesimal successive minima, then jets and finally polytopal upper and lower bounds on the generic iNOBodies. Moreover, in [FL25c] the authors study the shape of generic iNOBodies on Jacobians in small dimensions. The main issue here is realizing at least asymptotically the family $W_\bullet^t$ as *complete* graded linear series on a modification of $\mathbb{P}^{n-1}$. For example, for non-hyperelliptic Jacobian threefolds, $W_\bullet^t$ becomes complete on a birational model of $\mathbb{P}^2$ that is far from being toric, while for the hyperelliptic case we may need to go to the Zariski–Riemann space.

The paper is organized as follows. In Section 2 we review the Newton–Okounkov body construction and their basic properties. In Section 3 we use intersection theory, positivity, toric geometry and Euclidean volume computations to understand the vertical slices as Newton–Okounkov bodies on X_Σ . In Section 4 we treat the case of dimension $n \leq 3$. In Section 5 we use particular incomplete linear systems on X_Σ to prove that the vertices of the simplex from Theorem 1.1 do belong to the iNObody, concluding its proof. In Section 6 we state and motivate some open questions.